EXPERIMENT 3B – REGRESSION MODELS

AIM

To predict Sales from TV, Radio and Newspaper advertising budgets and compare Linear, Polynomial and RBF-SVR regression.

1. Load and inspect the Advertising dataset.
2. Select TV, Radio and Newspaper as X and Sales as y.
3. Plot pair plot and correlation heatmap.
4. Split data 80:20 and train three regressors.
5. Calculate MAE, MSE, RMSE and R^2; plot metric comparison and actual versus predicted sales.

In [ ]

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import pandas as pd, numpy as np, seaborn as sns, matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures,StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score

class="cm"># Load dataset
df=pd.read_csv(class="s">"Advertising.csv"); print(df.head(),df.tail(),df.shape

class="cm"># Adjust columns to match the actual dataset
features = [class="s">"Daily Time Spent on Site", class="s">"Age", class="s">"A
target = class="s">"Clicked on Ad"

X=df[features]; y=df[target]
sns.pairplot(df[features + [target]]); plt.show()
sns.heatmap(df[features + [target]].corr(),annot=True); plt.show()
print(class="s">"Most correlated:",df[features].corrwith(y).abs().idxmax())
Xtr,Xte,ytr,yte=train_test_split(X,y,test_size=.2,random_state=42)
models={class="s">"Linear":LinearRegression(),class="s">"Polynomial":make_pipel
rows=[]; preds={}
for n,m in models.items():
 m.fit(Xtr,ytr); p=m.predict(Xte); preds[n]=p; mse=mean_squared_error(yte,p)
 rows.append([n,mean_absolute_error(yte,p),mse,np.sqrt(mse),r2_score(yte,p)])
s=pd.DataFrame(rows,columns=[class="s">"Model",class="s">"MAE",class="s">"MSE",
s.set_index(class="s">"Model").plot(kind=class="s">"bar"); plt.tight_layout();
```

```
   Daily Time Spent on Site  Age  Area Income  Daily Internet Usage  \
0                     68.95   35     61833.90                256.09
1                     80.23   31     68441.85                193.77
2                     69.47   26     59785.94                236.50
3                     74.15   29     54806.18                245.89
4                     68.37   35     73889.99                225.58

                          Ad Topic Line            City  Male     Country
0     Cloned 5thgeneration orchestration     Wrightburgh     0     Tunisia
1     Monitored national standardization       West Jodi     1       Nauru
2       Organic bottom-line service-desk        Davidton     0  San Marino
3  Triple-buffered reciprocal time-frame  West Terrifurt     1       Italy
4          Robust logistical utilization    South Manuel     0     Iceland

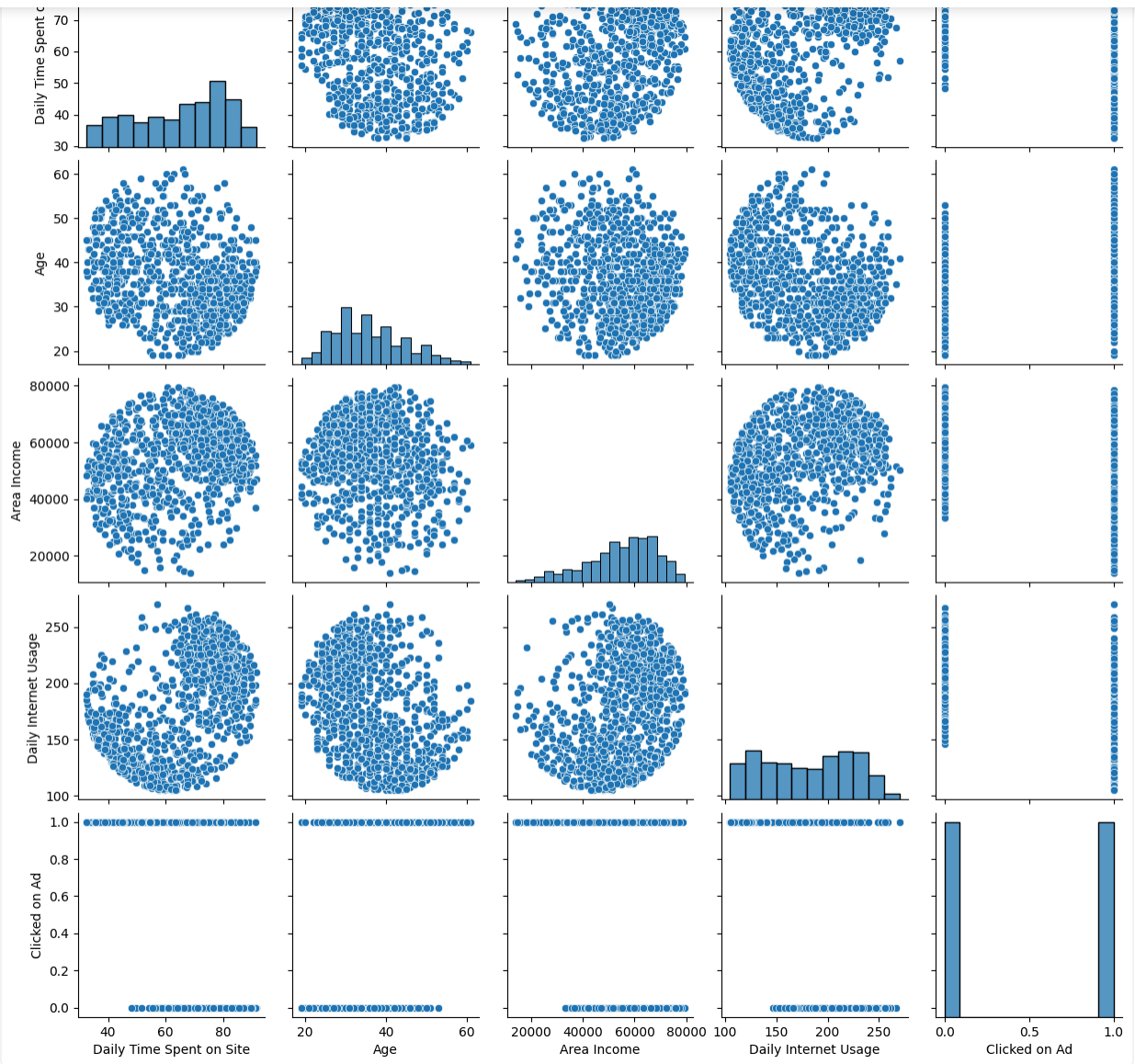
             Timestamp  Clicked on Ad
0  2016-03-27 00:53:11              0
1  2016-04-04 01:39:02              0
2  2016-03-13 20:35:42              0
3  2016-01-10 02:31:19              0
4  2016-06-03 03:36:18              0        Daily Time Spent on Site  Age  \
995                     72.97   30     71384.57                208.58
996                     51.30   45     67782.17                134.42
997                     51.63   51     42415.72                120.37
998                     55.55   19     41920.79                187.95
999                     45.01   26     29875.80                178.35

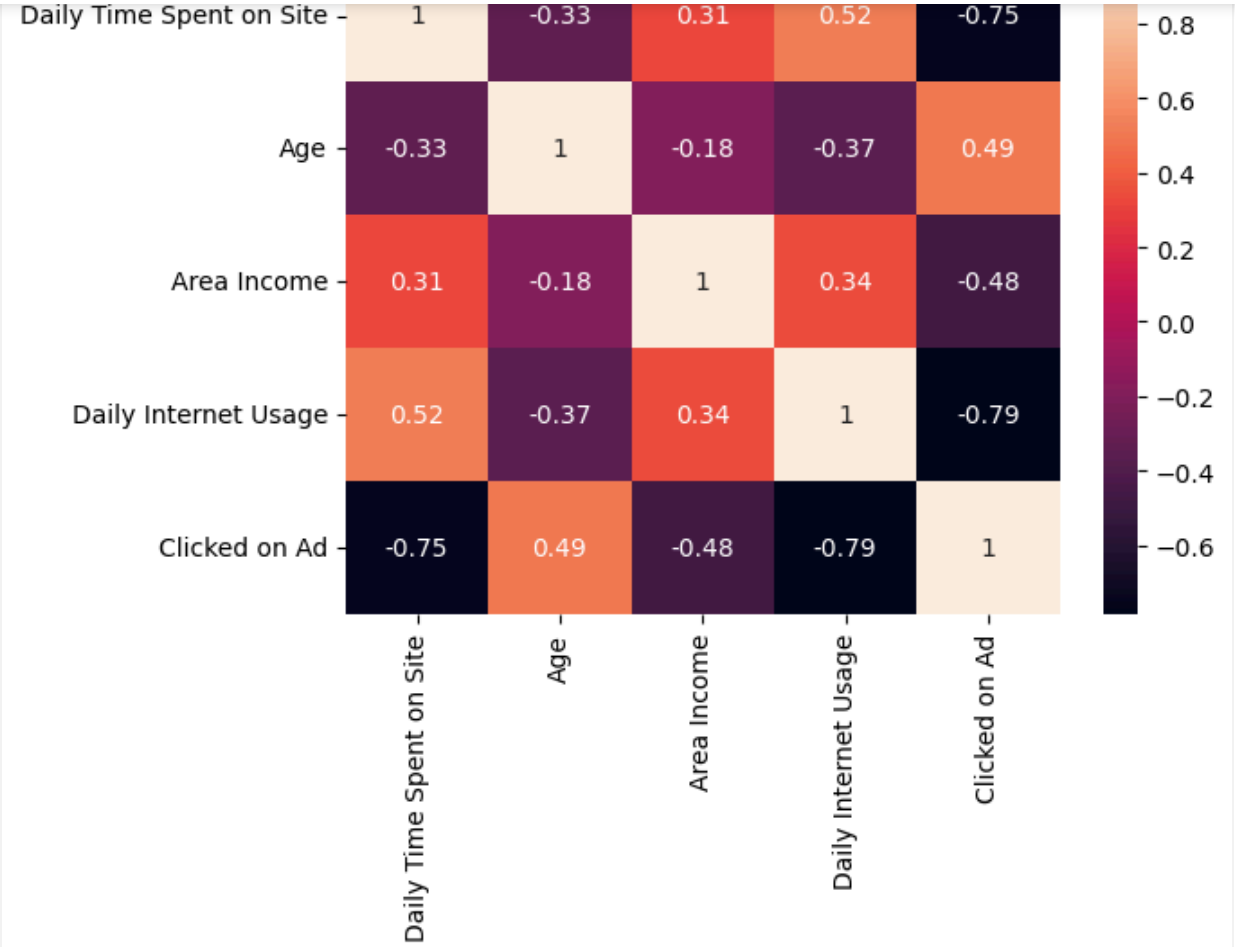
                          Ad Topic Line           City  Male  \
995         Fundamental modular algorithm      Duffystad     1
996       Grass-roots cohesive monitoring    New Darlene     1
997          Expanded intangible solution  South Jessica     1
998  Proactive bandwidth-monitored policy    West Steven     0
999       Virtual 5thgeneration emulation    Ronniemouth     0

                   Country            Timestamp  Clicked on Ad
995                Lebanon  2016-02-11 21:49:00              1
996  Bosnia and Herzegovina  2016-04-22 02:07:01              1
997               Mongolia  2016-02-01 17:24:57              1
998              Guatemala  2016-03-24 02:35:54              0
999                 Brazil  2016-06-03 21:43:21              1   (1000, 10)
```

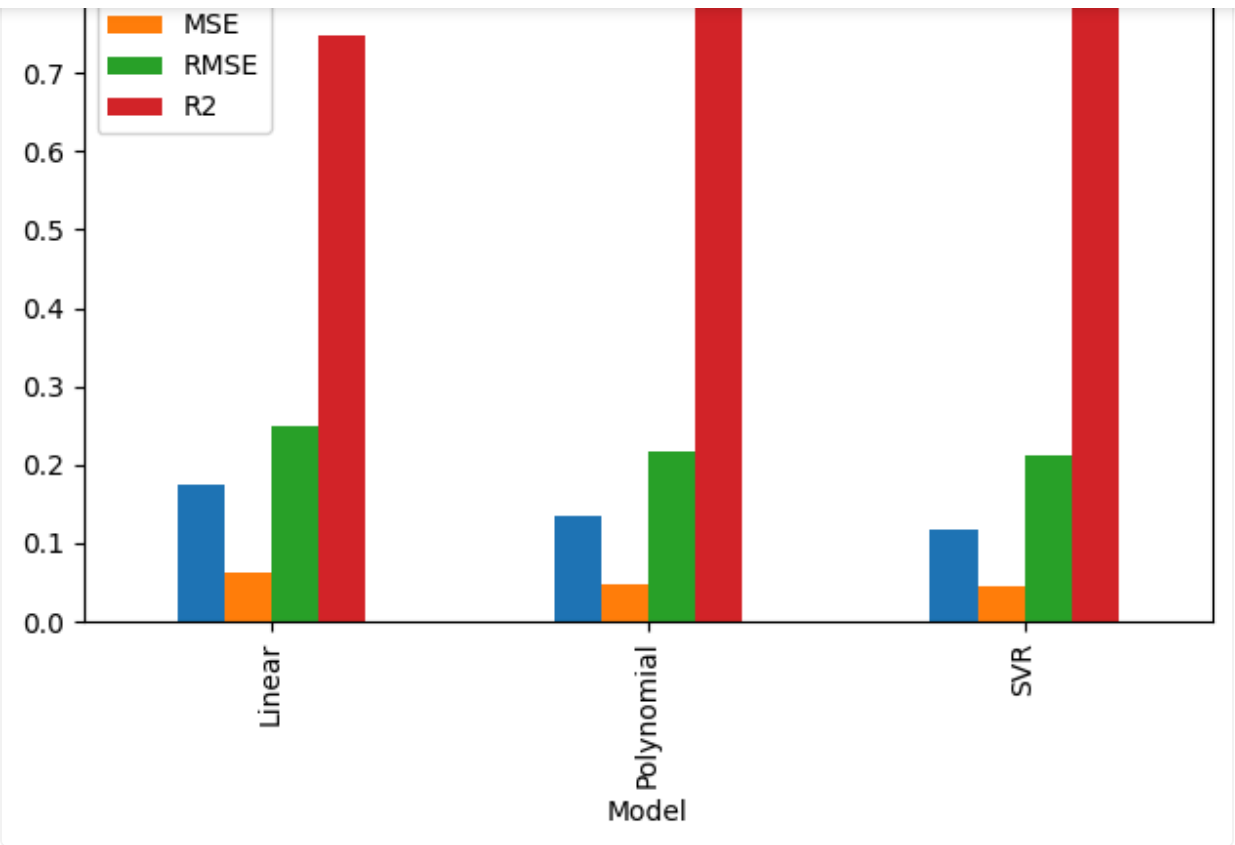
```
Ad Topic Line               0
City                        0
Male                        0
Country                     0
Timestamp                   0
Clicked on Ad               0
dtype: int64        Daily Time Spent on Site          Age   Area Income  \
count               1000.000000  1000.000000   1000.000000
mean                  65.000200    36.009000  55000.000080
std                   15.853615     8.785562  13414.634022
min                   32.600000    19.000000  13996.500000
25%                   51.360000    29.000000  47031.802500
50%                   68.215000    35.000000  57012.300000
75%                   78.547500    42.000000  65470.635000
max                   91.430000    61.000000  79484.800000

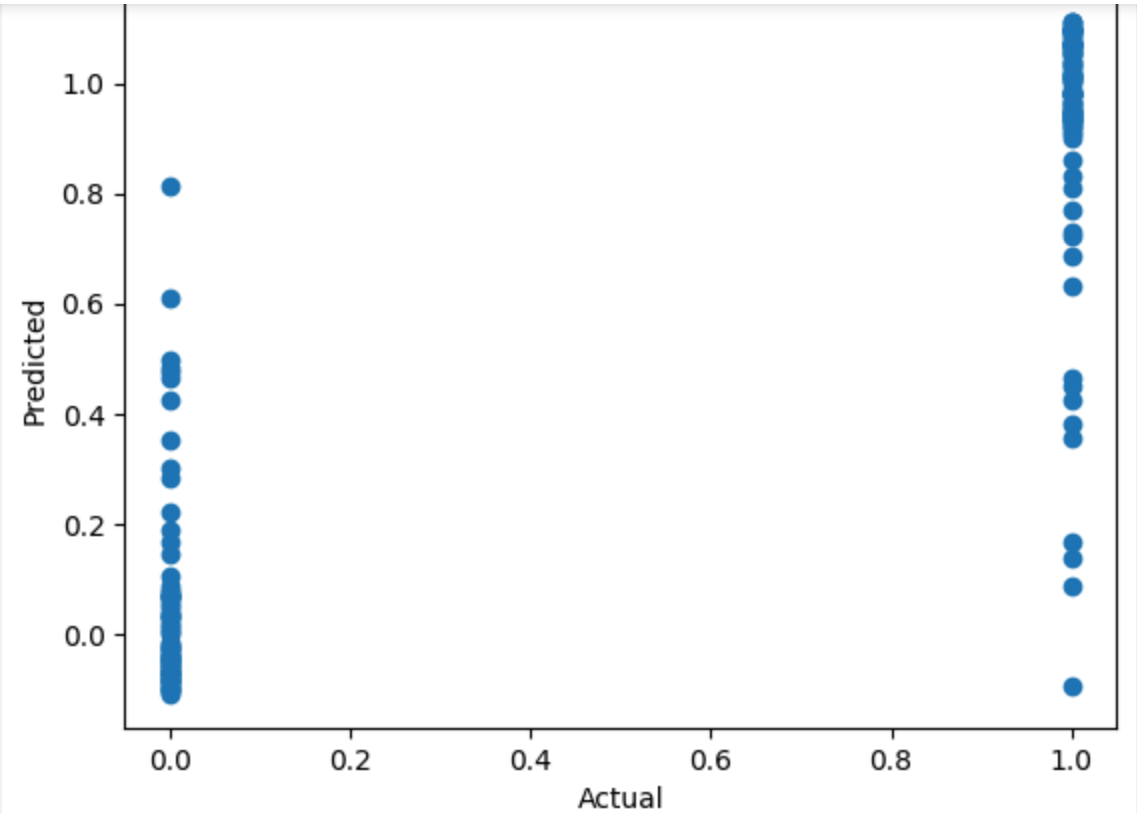
       Daily Internet Usage         Male  Clicked on Ad
count           1000.000000  1000.000000     1000.00000
mean             180.000100     0.481000        0.50000
std               43.902339     0.499889        0.50025
min              104.780000     0.000000        0.00000
25%              138.830000     0.000000        0.00000
50%              183.130000     0.000000        0.50000
75%              218.792500     1.000000        1.00000
max              269.960000     1.000000        1.00000
```


Daily Time Spent on Site
Age
Area Income
Daily Internet Usage
Clicked on Ad




```
Most correlated: Daily Internet Usage
        Model       MAE       MSE      RMSE        R2
0      Linear  0.174709  0.062366  0.249732  0.747480
1  Polynomial  0.135700  0.047560  0.218083  0.807429
2         SVR  0.117782  0.044758  0.211562  0.818773
```


MSE
RMSE
R2
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Linear
Polynomial
SVR
Model




```
Lowest RMSE: SVR
```

RESULT

Thus, regression models are compared and the model with the lowest prediction error is selected.